

## 1. General Description

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The EMS3905 is a dual SPDT low on-resistance analog switch. It can operate from 1.5V to 5.5V power supply. The device offers low ON-state resistance and excellent ON-state resistance matching with break-before-make feature, to prevent signal distortion during the transferring of a signal from one channel to another. The device is capable of truly isolation. Even when A overrides  $V_{CC}$ , very little current will flow back to the supply.

## 2. Features and Benefits

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- Wide Supply Range from 1.5 V to 5.5 V
- Low ON-state resistance, typically  $1.9\ \Omega$  ( $V_{CC}=5.5\text{ V}$ )
- 1.8 V Compatible Control Logic Inputs
- High Off-Isolation: -80 dB @ 10 kHz
- Low Channel-to-Channel Crosstalk: -82 dB @ 10 kHz
- High Bandwidth ( -3dB @ 800MHz) Suitable For USB2.0 High-Speed Routing
- Low Quiescent Current ( $< 1\mu\text{A}$ ) when  $V_{CC} = 3.3\text{ V}$

## 3. Applications

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- Audio, Video, UART, USB2.0 Signal and Supply Routing
- Cell phones and TWS headset

## EMS3905

Low Voltage and High Speed Dual SPDT Analog Switch with True Isolation

## 4. Ordering Information

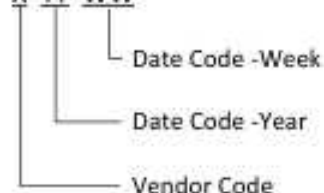
Table 1. Ordering information

| Type number | Topside marking | Package        |                                                              |          |
|-------------|-----------------|----------------|--------------------------------------------------------------|----------|
|             |                 | Name           | Description                                                  | Quantity |
| EMS3905MS   | S3905<br>XYYWW  | MSOP-10L       | MSOP package, 10 pins<br>3.0mm × 3.0mm; 1.1 mm (Max) height  | 3000     |
| EMS3905DRC  | S3905<br>XYYWW  | DFN3x3-10L     | DFN package, 10 pins<br>3.0 mm × 3.0 mm; 0.8 mm (Max) height | 5000     |
| EMS3905RSW  | 3905<br>YYWW    | QFN1.4x1.8-10L | QFN package, 10 pins<br>1.4 mm × 1.8 mm; 0.6 mm (Max) height | 3000     |

### MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Date Code.

X YY WW



## 5. Pinning Information

### 5.1. Pin map

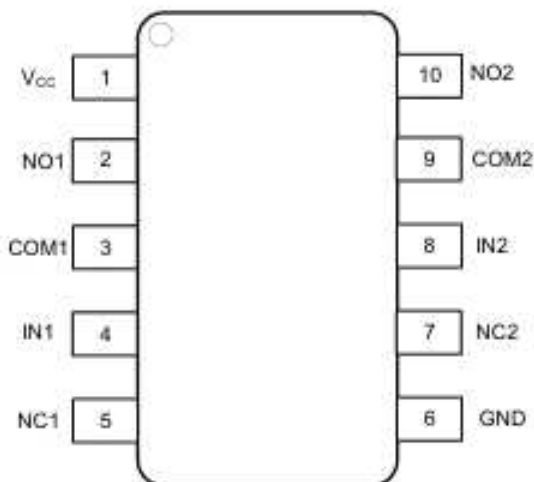


Fig 1. Top view pin configuration MSOP-10L

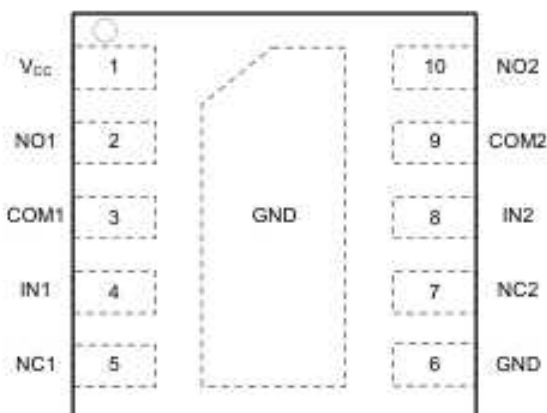


Fig 2. Top view pin configuration DFN3x3-10L

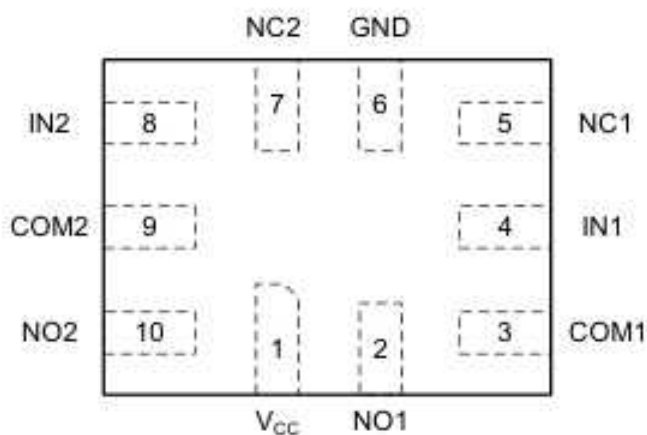


Fig 3. Top view pin configuration QFN1.4x1.8-10L

## EMS3905

### Low Voltage and High Speed Dual SPDT Analog Switch with True Isolation

## 5.2. Pin description

Table 2. Pin description

| Symbol | Pin                                 | Description                      |
|--------|-------------------------------------|----------------------------------|
|        | MSOP-10L/DFN3x3-10L/ QFN1.4x1.8-10L |                                  |
| VCC    | 1                                   | Power supply pin                 |
| NO1    | 2                                   | Normally open I/O for switch 1   |
| COM1   | 3                                   | Common I/O for switch 1          |
| IN1    | 4                                   | Select pin for switch 1          |
| NC1    | 5                                   | Normally closed I/O for switch 1 |
| GND    | 6                                   | Ground                           |
| NC2    | 7                                   | Normally closed I/O for switch 2 |
| IN2    | 8                                   | Select pin for switch 2          |
| COM2   | 9                                   | Common I/O for switch 2          |
| NO2    | 10                                  | Normally open I/O for switch 2   |

## 6.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

| IN | NC TO COM,<br>COM TO NC | NO TO COM,<br>COM TO NO |
|----|-------------------------|-------------------------|
| L  | ON                      | OFF                     |
| H  | OFF                     | ON                      |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol          | Parameter                                                        | Conditions | Min  | Max       | Unit |
|-----------------|------------------------------------------------------------------|------------|------|-----------|------|
| $V_{CC}$        | supply voltage                                                   |            | -0.3 | 6.5       | V    |
| $V_{IN}$        | control Input Voltage                                            |            | -0.3 | 6.5       | V    |
| $V_A$           | signal input voltage                                             |            | -0.3 | 6.5       | V    |
| $I_A$           | continuous Current Through COM, NC, NO                           |            |      | $\pm 100$ | mA   |
|                 | peak Current Through COM, NC, NO (pulsed at 1ms, 50% duty cycle) |            |      | $\pm 200$ | mA   |
| $T_J$           | junction Temperature under Bias                                  |            |      | 150       | °C   |
| $T_L$           | lead Temperature (Soldering, 10 seconds)                         |            |      | 260       | °C   |
| $R_{\theta JA}$ | thermal resistance                                               |            |      | 350       | °C/W |

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol    | Parameter             | Min | Typ | Max      | Unit |
|-----------|-----------------------|-----|-----|----------|------|
| $V_{CC}$  | supply voltage        | 1.5 | 3.3 | 5.5      | V    |
| $V_{IN}$  | control input voltage | 0   |     | 5.5      | V    |
| $V_A$     | signal input voltage  | 0   |     | $V_{CC}$ | V    |
| $T_{amb}$ | ambient temperature   | -40 | 25  | 125      | °C   |

## 9. Electrical Characteristics

**Table 6. Static characteristics**
 $T_{amb}=25^{\circ}\text{C}$ ,  $V_{CC}=3.3\text{ V}$ , unless otherwise specified

| Symbol                               | Parameter                                               | Conditions                                                                               | Min | Typ  | Max       | Unit          |
|--------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------|-----|------|-----------|---------------|
| <b>DC Electrical Characteristics</b> |                                                         |                                                                                          |     |      |           |               |
| $V_{IH}$                             | input logic high level                                  | $V_{CC} = 3.4\text{ V to }5.5\text{ V}$                                                  | 1.6 |      |           | V             |
|                                      |                                                         | $V_{CC} = 1.5\text{ V to }3.3\text{ V}$                                                  | 1.4 |      |           | V             |
| $V_{IL}$                             | input logic low level                                   | $V_{CC} = 3.4\text{ V to }5.5\text{ V}$                                                  |     |      | 0.6       | V             |
|                                      |                                                         | $V_{CC} = 1.5\text{ V to }3.3\text{ V}$                                                  |     |      | 0.4       | V             |
| $I_{CC}$                             | supply current                                          | $I_A = 0$ , $V_{SEL} = 0$ or $V_{SEL} = V_{CC}$                                          |     |      | 1.0       | $\mu\text{A}$ |
| $\Delta I_{CC}$                      | additional supply current                               | $I_A = 0$ , $V_{CC} = 2.5\text{ V or }4.2\text{ V}$<br>$V_{SEL} = 1.6\text{ V}$          |     |      | 3.0       | $\mu\text{A}$ |
| $I_{off}$                            | OFF-state switch leakage current from COM to NC (or NO) | $V_A = 5\text{ V}$ , $V_{NC}$ or $V_{NO} = 0\text{ V}$                                   |     |      | $\pm 2.0$ | $\mu\text{A}$ |
| $R_{ON}$                             | on-resistance                                           | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 3.4  | 4.0       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 2.4  | 3.0       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 1.9  | 2.5       | $\Omega$      |
| $R_{FLAT}$                           | on-resistance flatness                                  | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.2  | 0.5       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.25 | 0.5       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.3  | 0.5       | $\Omega$      |
| $\Delta R_{ON}$                      | on-resistance matching between channels                 | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |

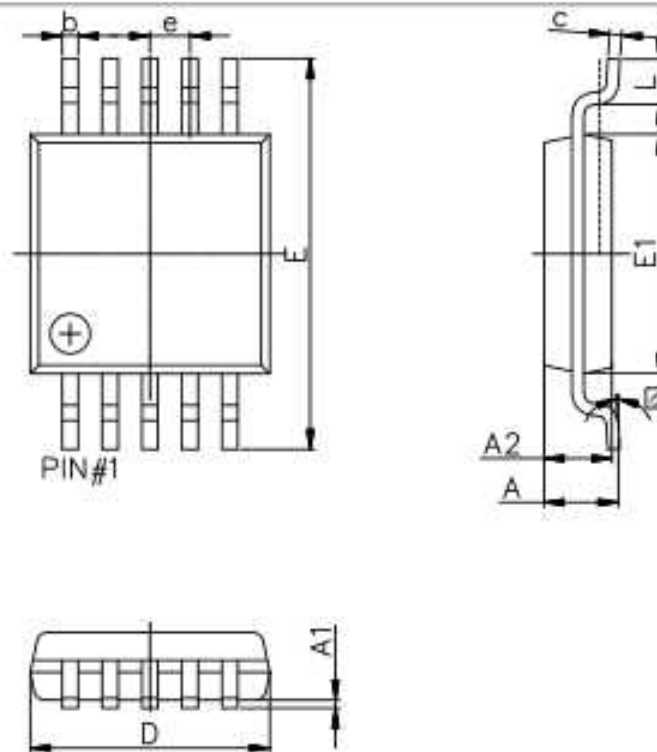
## EMS3905

### Low Voltage and High Speed Dual SPDT Analog Switch with True Isolation

| Symbol                                | Parameter                 | Conditions                                                                | Min | Typ | Max | Unit |
|---------------------------------------|---------------------------|---------------------------------------------------------------------------|-----|-----|-----|------|
| <b>AC Electronics Characteristics</b> |                           |                                                                           |     |     |     |      |
| $t_{ON}$                              | turn-On Time              | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                                  |     | 470 |     | ns   |
| $t_{OFF}$                             | turn-Off Time             | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                                  |     | 120 |     | ns   |
| $t_{BBM}$                             | break-before-make time    | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                                  |     | 600 |     | ns   |
| BW                                    | -3dB Bandwidth            | $C_L=0pF$                                                                 |     | 800 |     | MHz  |
| OIRR                                  | off isolation             | $f=10\text{ KHz}$                                                         |     | -80 |     | dB   |
|                                       |                           | $f=240\text{ MHz}$                                                        |     | -31 |     | dB   |
| $X_{TALK}$                            | crosstalk                 | $f=10\text{ KHz}$                                                         |     | -82 |     | dB   |
|                                       |                           | $f=240\text{ MHz}$                                                        |     | -26 |     | dB   |
| THD                                   | total harmonic distortion | $f=20\text{Hz to } 20\text{KHz}$<br>$V_A=600\text{mVpp}$ , $R_L=32\Omega$ |     | -80 |     | dB   |
| <b>Capacitance</b>                    |                           |                                                                           |     |     |     |      |
| $C_{ON}$                              | On Capacitance            | $f=100\text{KHz}$                                                         |     | 12  |     | pF   |
| $C_{OFF}$                             | Off Capacitance           | $f=100\text{KHz}$                                                         |     | 4   |     | pF   |

## 10. Package Outline

### MSOP-10L

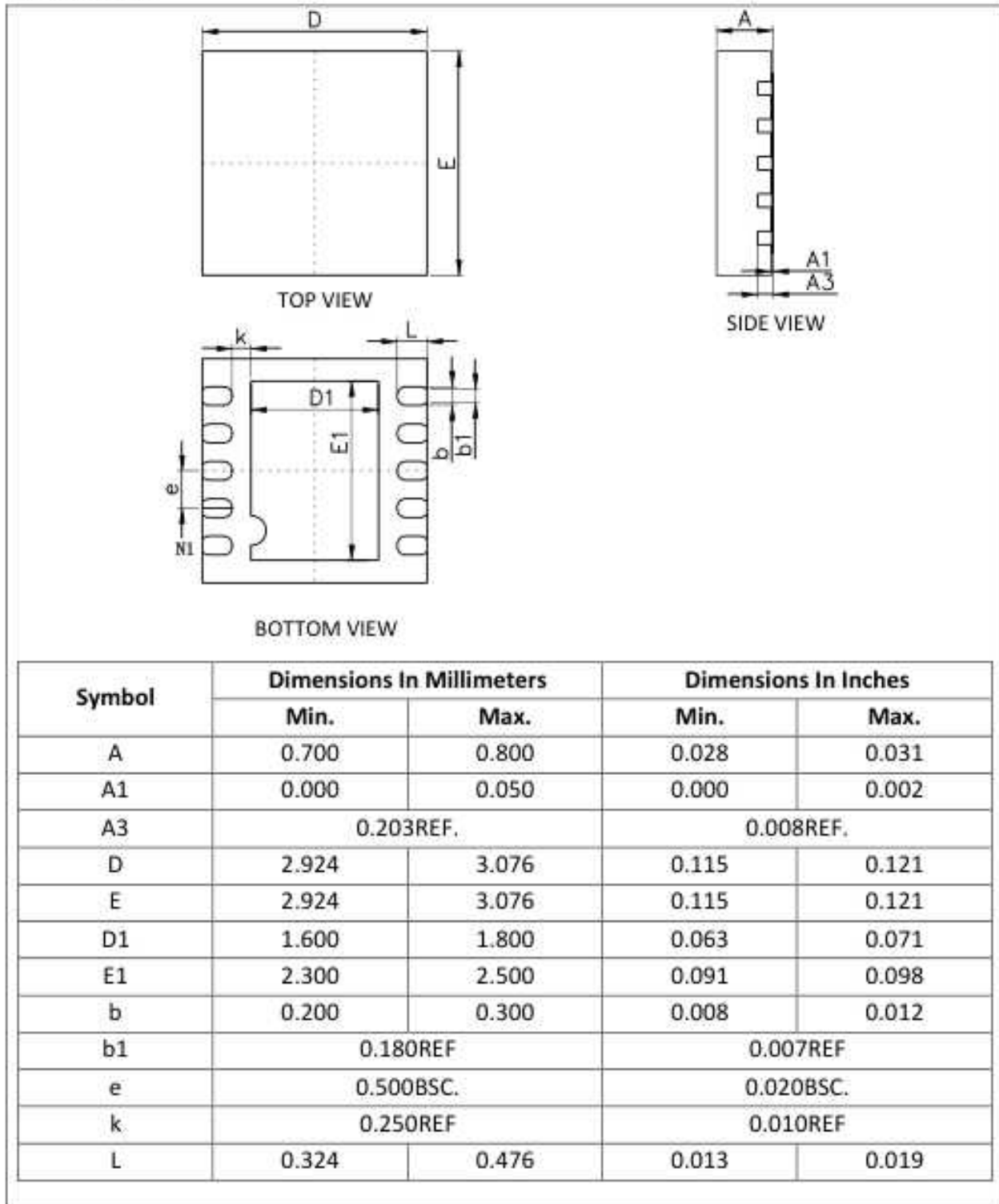


| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| A      | ---                       | 1.100 | ---                  | 0.043 |
| A1     | 0.020                     | 0.150 | 0.001                | 0.006 |
| A2     | 0.750                     | 0.950 | 0.030                | 0.037 |
| b      | 0.180                     | 0.330 | 0.007                | 0.013 |
| c      | 0.090                     | 0.230 | 0.004                | 0.009 |
| D      | 2.900                     | 3.100 | 0.114                | 0.122 |
| e      | 0.500(BSC)                |       | 0.020(BSC)           |       |
| E      | 4.750                     | 5.050 | 0.187                | 0.199 |
| E1     | 2.900                     | 3.100 | 0.114                | 0.122 |
| L      | 0.400                     | 0.800 | 0.016                | 0.031 |
| θ      | 0°                        | 6°    | 0°                   | 6°    |

# EMS3905

## Low Voltage and High Speed Dual SPDT Analog Switch with True Isolation

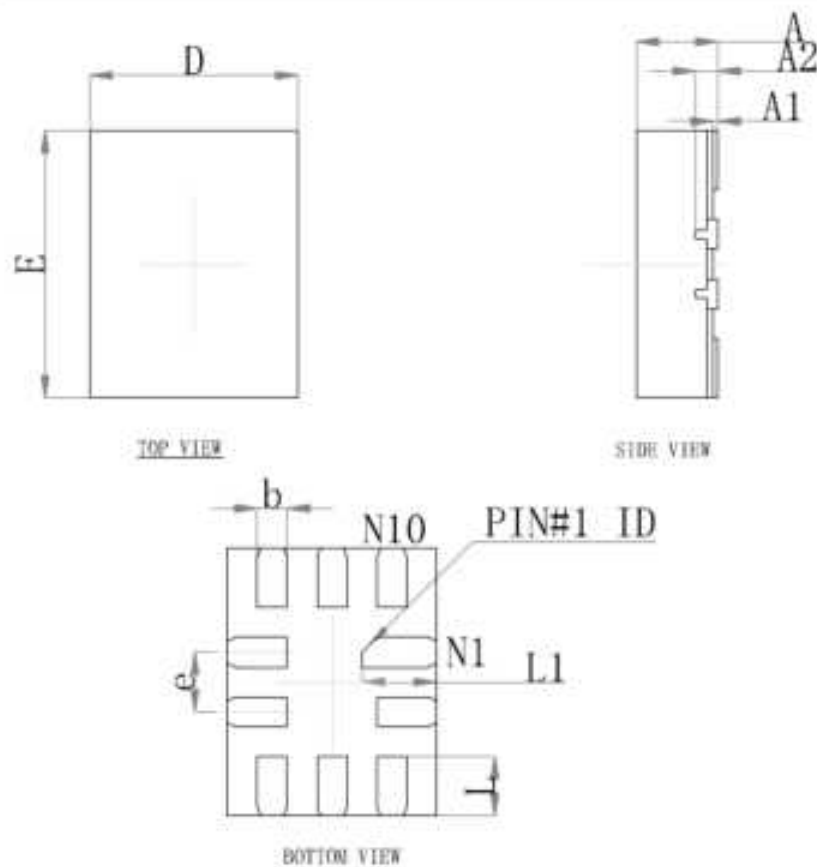
### DFN 3x3 -10L



# EMS3905

## Low Voltage and High Speed Dual SPDT Analog Switch with True Isolation

### QFN 1.4x1.8 -10L



| Symbol | Dimension in Millimeters |       |       |
|--------|--------------------------|-------|-------|
|        | Min.                     | NOM.  | Max.  |
| A      | 0.500                    | 0.550 | 0.600 |
| A1     | 0.000                    | --    | 0.050 |
| A2     | 0.152 Ref.               |       |       |
| D      | 1.324                    | 1.400 | 1.476 |
| E      | 1.724                    | 1.800 | 1.876 |
| b      | 0.150                    | 0.200 | 0.250 |
| e      | 0.400 Typ.               |       |       |
| L      | 0.350                    | 0.400 | 0.450 |
| L1     | 0.450                    | 0.500 | 0.550 |

## 11. Tape and Reel Information

### 11.1. Carrier tape dimensions

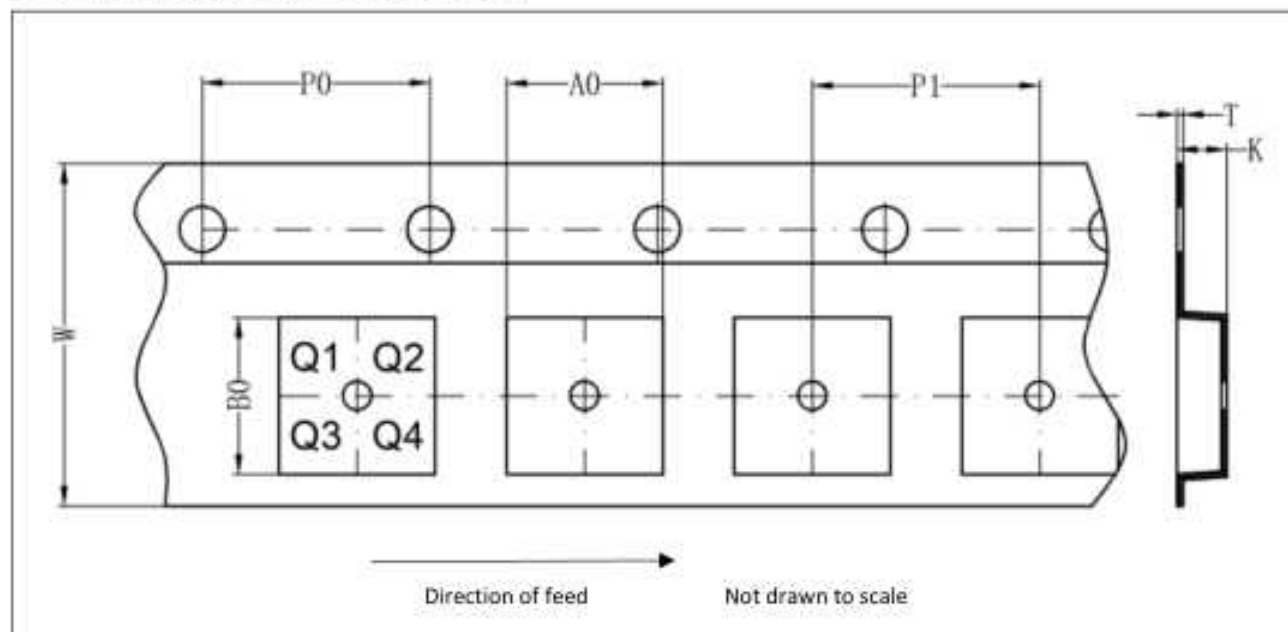
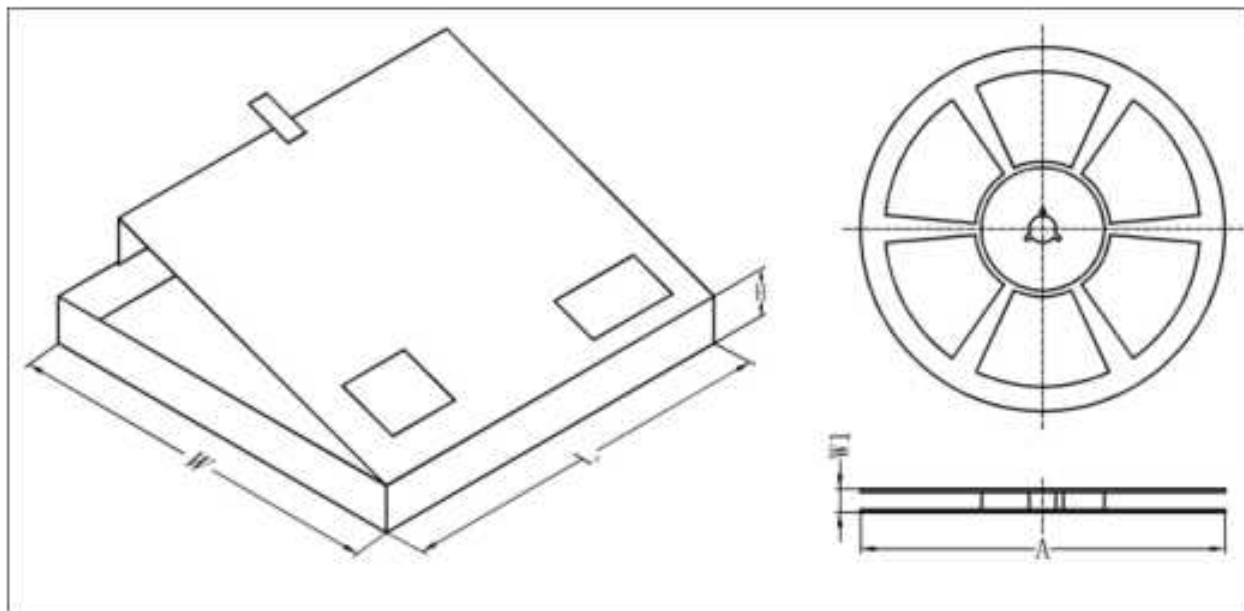


Table 10. Carrier tape dimensions

| Package version | A0(mm) | B0(mm) | K(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|-------|-------|--------|-------|--------|-------|
| MSOP-10L        | 5.3    | 3.4    | 1.4   | 0.25  | 8      | 12    | 4      | Q1    |
| DFN3x3-10L      | 3.35   | 3.35   | 1.13  | 0.25  | 8      | 12    | 4      | Q1    |
| QFN1.4x1.8-10L  | 1.6    | 2.0    | 0.85  | 0.2   | 4      | 8     | 4      | Q1    |

## 11.2. Reel and box dimensions



**Table 11. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Max Reel Width W1 (mm) | SPQ (pcs)[1] | Reels per box | Outer box dimensions L×W×H(mm)[2] |
|-----------------|-----------------|-----------------------|------------------------|--------------|---------------|-----------------------------------|
| MSOP-10L        | MS              | 330                   | 18.4                   | 3000         | 1             | 358x340x50                        |
| DFN3x3-10L      | DRC             | 330                   | 18.4                   | 5000         | 1             | 358x340x50                        |
| QFN1.4×1.8-10L  | RSW             | 180                   | 13.2                   | 3000         | 1             | 210x200x40                        |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 12. Revision History

Table 7. Revision history

| Document ID      | Release Date                                                                                                                                  | Data sheet status | Change notice | Supersedes       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|------------------|
| EMS3905 Rev. 1.3 | Apr 15, 2026                                                                                                                                  | Product datasheet |               | EMS3905 Rev. 1.2 |
| Modifications:   | <ul style="list-style-type: none"> <li>Table 1 ordering information updated.</li> <li>Section 11: Added Tape and Reel Information.</li> </ul> |                   |               |                  |
| EMS3905 Rev. 1.2 | Jun 10, 2024                                                                                                                                  | Product datasheet |               | EMS3905 Rev. 1.1 |
| Modifications:   | <ul style="list-style-type: none"> <li>Table2 updated.</li> <li>Table6: <math>R_{FLAT}</math> updated.</li> </ul>                             |                   |               |                  |
| EMS3905 Rev. 1.1 | Apr 10, 2024                                                                                                                                  | Product datasheet |               | EMS3905 Rev. 1.0 |
| Modifications:   | <ul style="list-style-type: none"> <li>Added the QFN package outline.</li> </ul>                                                              |                   |               |                  |
| EMS3905 Rev. 1.0 | Feb 13, 2024                                                                                                                                  | Product datasheet |               |                  |